**SPECIFICATION**

# SYSTEM AND METHOD FOR HIGH-RADIX NETWORK COMMUNICATION, DYNAMIC LEXICON ROTATION, AND REAL-TIME IMMUTABLE ARCHIVAL

## Cross-Reference to Related Applications

This application is related to and incorporates by reference the subject matter of the following co-pending U.S. Patent Applications in their entirety:

- U.S. Patent Application No. 19/560,031, titled "METHOD AND SYSTEM FOR PIXEL-BASED NETWORK ROUTING IN A MASSIVELY PARALLEL OPTICAL INTERCONNECT";
- U.S. Patent Application No. 19/562,825, titled "SYSTEM FOR AN OPTICALLY-NATIVE CENTRAL PROCESSING UNIT";
- U.S. Patent Application No. 19/639,271, titled "High-Radix Mass Storage System";
- U.S. Patent Application No. 19/639,265, titled "A High-Radix and Parametric Satellite Communication System";
- U.S. Patent Application No. 19/639,255, titled "High-Radix and Parametric Cellular Communication System";
- U.S. Patent Application No. 19/639,258, titled "High-Radix and Parametric Point-to-Point RF Communication System";
- U.S. Patent Application No. 19/639,253, titled "A Protocol Stack and System for Parametric Data Networking";

- U.S. Patent Application No. 19/764,830, titled "HYBRID HIGH-RADIX PAPER-DIGITAL RECORD ARCHIVAL AND IMMUTABLE LEDGER SYSTEM";
- U.S. Patent Application No. 19/561,259, titled "PROTOCOL FOR ENCODING MAP ROTATION FOR FORWARD SECRECY";
- U.S. Patent Application No. 19/561,223, titled "SYSTEM AND METHOD FOR N-GRAM PARALLEL DATA ENCODING"; and
- U.S. Patent Application No. 19/640,973, titled "SECURE PARAMETRIC NETWORK PROTOCOL WITH ROTATING ENCODING MAPS FOR FORWARD SECRECY".

## Background of the Invention

Modern communication networks predominantly operate under serialized binary architectures where information is encoded into standardized byte streams. Fundamentally, this infrastructure is a serialized text-based system of communication. Because the lexicon of communication is static—relying on stable grammars, consistent character spellings, and predictable protocol headers—adversaries can intercept and eavesdrop on network traffic simply by performing frequency analysis on predictable data streams.

While conventional cryptography encrypts the contents of these streams, the underlying distributions of ciphertext blocks, metadata footprints, and protocol exchanges remain tied to static serialized structures. Furthermore, as communication bandwidth demands scale into the multi-terabit and petabit regimes, legacy serial-binary hardware hits insurmountable thermal, spatial, and memory walls imposed by electrical wiring and serial conversion bottlenecks. While emerging high-radix optical systems offer orders-of-magnitude increases in throughput, legacy systems down-convert optical signals into serialized bytes, squandering the physical bandwidth advantages. There is an urgent need for an end-to-end high-radix network architecture that natively communicates, routes, and processes information in multi-spectral optical states, while offering backward-compatible retrofitting for legacy binary infrastructure.

## Summary of the Invention

The present invention provides a system and method for a high-radix network architecture that natively operates on high-radix multi-spectral visual state frames rather than serialized byte streams, incorporating forward-secret rotating lexicons and enabling real-time immutable physical archival.

In a preferred embodiment, the system comprises an end-to-end natively optical high-radix network. In this native architecture, data is natively originated, processed by optically-native central processing units (such as described in U.S. Patent Application No. 19/562,825), routed across optical networks via pixel-based routing matrices (such as described in U.S. Patent Application No. 19/560,031), and communicated across terrestrial, satellite, cellular, and point-to-point channels using a unified parametric protocol stack (such as described in U.S. Patent Application Nos. 19/639,253, 19/639,265, 19/639,255, and 19/639,258). Throughout transmission, the semantic meaning of the visual states (colors, pixels, and n-grams) dynamically and systematically rotates based on forward-secret parametric maps unique to each communicating pair of nodes.

In a transitional backwards-compatible embodiment, the system retrofits legacy serial-binary computers and existing internet infrastructure. A software or hardware codec translates standard serialized byte streams into high-radix visual frames and vice versa. Although legacy hardware bottlenecks line-rate optical bandwidth gains, the system immediately achieves absolute cryptanalytic privacy over legacy networks: by dynamically rotating the visual lexicons, the statistical significance of predictable linguistic, socio-economic, and behavioral footprints is eliminated across the entire network.

Additionally, the invention leverages the extreme physical information density of high-radix optical frames to achieve real-time immutable physical archival. In secure environments (e.g., intelligence centers or situation rooms), massive line-rate optical network streams are dumped directly onto high-density physical optical media (such as high-radix WORM matrices) without executing a visual-to-binary down-conversion. This eliminates traditional storage write-speed bottlenecks and thermal limits, providing a permanent, tamper-proof audit trail of network traffic.

## Statement of Conditional Sovereign License Grant

To incentivize the rapid deployment of this technology, the Inventor hereby grants a perpetual, royalty-free, non-exclusive license to the United States Government and its defense and civil procurement agencies to practice and use this invention worldwide for any and all governmental purposes.

However, this public sovereign license grant is subject to a strict, mandatory legal pre-condition: The license shall only vest and become active after the technology described in USPTO Non-Provisional Patent Application No. 19/764,830 has been successfully deployed and implemented to secure the state judiciary court records systems of the State of Texas, the State of Florida, and the State of Washington. The implementation in the state court records systems of these three states shall be fully covered under the perpetual royalty-free license. Upon successful implementation in these three target states, the royalty-free license to the United States Government automatically expands to be worldwide. This structure legally incentivizes the federal government to apply administrative pressure and funding to these three states to adopt this fraud-proof technology disclosed in USPTO Application No. 19/764,830.

## Brief Description of the Drawings

FIG. 1 is a block diagram illustrating the end-to-end natively high-radix network architecture and transitional backward-compatible retrofitting interface.

FIG. 2 is a block diagram illustrating the real-time immutable optical archival system operating on native high-radix network streams.

## Detailed Description of the Invention

Referring to FIG. 1, an architectural block diagram of the high-radix network communication system is illustrated. In the preferred embodiment, the system operates as an end-to-end native optical

network. A transmitting native optical terminal (100) comprises an optically-native central processing unit (102) (such as described in U.S. Patent Application No. 19/562,825) that generates wide data words directly in native multi-spectral optical states without binary down-conversion. A native parametric lexicon engine (104) dynamically rotates the semantic visual mapping between communicative states and multi-spectral pixel arrays in accordance with a forward-secret parametric protocol. The resulting high-radix visual frame (106) is transmitted into a high-radix optical network (110). Within the network, intermediate routing is handled by pixel-based network routers (112) (such as described in U.S. Patent Application No. 19/560,031) that inspect and switch visual frames natively on a parallel optical interconnect without serialized deserialization. At the destination terminal (120), a native optical receiver (122) decodes the frames utilizing a synchronized parametric lexicon engine (124) to deliver native optical payloads directly to destination optically-native processors (126).

FIG. 1 also illustrates the transitional backwards-compatible embodiment for legacy serial-binary environments. A legacy computing device (130) generates a conventional serialized binary byte stream (132). A software or hardware retrofit codec (134) (which may be embodied in an application such as a high-radix web browser or network driver) ingests the binary byte stream and maps the bytes to high-radix colored pixels or n-grams under the direction of the rotating parametric lexicon engine (104). The encoded visual frame is then encapsulated and transmitted across legacy network channels (136) (such as the legacy Internet). At the receiving end, a complementary retrofit codec (138) decodes the pixels back into a legacy byte stream (140) for a legacy destination computer (142). Although line-rate optical bandwidth is constrained by the legacy conversion hardware, this transitional mode eliminates predictable plaintext and ciphertext distributions across the public network, achieving forward secrecy and immunity from mass eavesdropping.

Referring to FIG. 2, the real-time immutable archival system is illustrated. A high-throughput optical network trunk (200) carries continuous line-rate high-radix data frames (202). In a high-security facility or operations control center (210), an optical interface tap (212) directs the native optical stream into an optical buffer array (214). An immutable physical archival engine (220) comprises an optical matrix recording head (222) that executes a direct native optical dump of the multi-spectral pixel states onto a high-density immutable physical optical media substrate (226)

(such as described in U.S. Patent Application Nos. 19/764,830 and 19/639,271). This native optical dump completely bypasses the write-speed, thermal, and spatial bottlenecks of legacy serial-binary conversion interfaces (228), enabling permanent, unalterable physical recording of massive network traffic in real time.